

RiskRadar — ML / Data Engineering Progress Update

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Author: Ire (ML / Data Engineer) **Covers:** PRD reconciliation, leakage removal, model swap, held-out evaluation, unseen-typology test **Status:** Third addition — calibration and alert-budget decision to follow next

1. Overview

This update covers reconciling our ML work against the team's finalized PRD (v2.0), fixing a data-leakage issue the PRD explicitly forbids, retraining on a cleaner feature set, and running the two evaluations the PRD requires: a standard held-out test and an unseen-typology test.

2. PRD reconciliation

The finalized PRD (§9) sets requirements our earlier work did not fully meet:

- **No pre-computed "risk/score" columns as features** — training on a column someone else already labeled with a risk score is target leakage.
- **~15 features, organised and explainable**, not an unbounded feature list.
- **Three named fraud patterns:** account compromise, money-mule fan-out, card testing.
- **Report Recall, Precision, false-positive burden — never accuracy** as the headline metric (at 3.6% fraud, "predict nothing is fraud" already scores 96.4%, which is meaningless).
- **Run an unseen-typology test:** train with one fraud pattern held out entirely, then test recall specifically on that withheld pattern.

We also received a "deliverables" status document describing a fully custom-built bank simulator, a calibrated gradient-boosted model, and completed public-dataset validation. On inspection, **none of the specific files, scripts, or results in that document exist in this repository** — it describes a different, further-along implementation than what has actually been built. This was flagged and the team confirmed: **continue with the current HuggingFace-based dataset**, and the PRD will be reviewed to accommodate that choice rather than requiring a from-scratch simulator.

Pattern mapping — a disclosed limitation

PRD pattern	Our dataset's closest match	Real row count
Account compromise	Account Takeover	~173,730
Money-mule fan-out	No match exists	—
Card testing	No match exists	—
(no PRD equivalent)	Identity Fraud	~9,312
(no PRD equivalent)	Impossible Travel Fraud	12

Money-mule fan-out and card testing do not exist anywhere in this dataset. This is disclosed here rather than worked around silently.

3. Leakage fix

Four columns were removed from the feature set before any further training: `merchant_fraud_rate` , `channel_risk_score` , `persona_fraud_risk` , `location_fraud_risk` . These are pre-computed summaries of how fraud-prone a category already is — in the previous baseline, two of them ranked in the top 3 features by importance, meaning the model was partly learning "this category was already tagged risky" rather than genuine behavioural patterns. This inflates reported performance and would not function in a live system anyway, since no such score exists at real transaction time.

4. Model swap

Replaced `RandomForestClassifier` with `HistGradientBoostingClassifier` :

- Matches the model family named in the team's reference deliverables document.
 - Meaningfully more memory-efficient on large data — training on 800,000 rows took **7.3 seconds**, versus several minutes and repeated `MemoryError` s with Random Forest on this machine (7.5GB RAM).
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5. Held-out evaluation (the core deliverable)

Trained on an 800,000-row stratified sample (25 features, leaky columns excluded), evaluated on a 200,000-row held-out test set never touched during training.

Default threshold (0.5) — reported for the record, not used going forward:

Metric	Value
Recall (fraud)	94.9%
Precision (fraud)	4.4%
False-positive burden	771 per 1,000 legitimate transactions

Threshold sweep — the real, usable result:

Threshold	Precision	Recall	FP burden /1,000
0.3	4.4%	99.9%	813.1
0.5	4.4%	94.9%	771.1
0.6	5.2%	0.7%	4.5
0.7	8.6%	0.2%	0.7

Finding: there is a hard cliff between thresholds 0.5 and 0.6 — no usable middle ground exists. Below the cliff, recall is excellent but the false positive burden is unworkable for a real analyst queue; above it, recall collapses to near-zero. This ceiling was already observed with the earlier Random Forest attempt; reproducing it here — with leaky features removed and a different model type — confirms it is a genuine property of this dataset, not a modelling mistake or a leakage artifact.

Implication: the ML score cannot drive decisions alone. This is the direct evidence for the PRD's hybrid design (§8.6–8.7) — ML probability and rule findings must stay separate, with the rule engine narrowing the alert volume the ML component would otherwise produce on its own.

6. Unseen-typology test

Per PRD §9.5: trained the model with each fraud type entirely removed from training data, then measured recall purely on the withheld examples.

Fraud type	Held-out examples	Recall when never trained on it
Account Takeover	34,179	61.4%
Identity Fraud	1,812	95.75%
Impossible Travel Fraud	—	Skipped — only 12 total examples exist in the dataset; not enough to report an honest figure

Finding: generalisation is uneven across pattern types. Identity Fraud's behavioural signature is well captured by features unrelated to its specific label (the model catches it even having never seen an example). Account Takeover relies more heavily on having seen the pattern directly — likely because its dominant signal (`is_ato_risk` / new-device transactions) is close to type-specific, so a model that never saw this label loses much of that signal's value.

7. Summary table

Task	Status
Reconcile ML approach with finalized PRD	Done — limitations disclosed rather than hidden
Remove leaked features	Done
Retrain on cleaned feature set	Done
Held-out evaluation (recall/precision/FP burden)	Done
Unseen-typology test	Done (2 of 3 patterns; 1 skipped for insufficient data)

8. What's next (fourth addition)

- **Calibration:** wrap the model so its stated fraud probability matches its actual accuracy where alerts are raised (probability honesty, calibration error).
- **Alert-budget decision:** bring the threshold-sweep table to the team so Chidera/Kanyinsola can pick an operating point balancing recall against realistic analyst workload, per PRD §8.9.
- **Optional stretch:** PaySim public-dataset validation, only if time allows after the above — not required by the PRD, and IEEE-CIS validation is deprioritised entirely given the effort it would require versus what it would prove.